Abdullah Bakr

AI / Machine Learning Engineer

+20 1553996442 | abdubakora1232@gmail.com | linkedin.com/in/abdubakr | github.com/abdubakr77 | kaggle.com/abdullahbakr7

Profile

AI/ML and Deep Learning Engineer with 3+ years of Python experience in computer vision, transfer learning, and end-to-end ML pipelines. Built 7+ real-world projects, including medical imaging AI, multiclass image classifiers, and an open-source data cleaning library on PyPI with 13,000+ downloads. Currently completing a remote ML Engineering internship at FlyRank AI, selected from 20,000+ applicants. Seeking AI/ML internships and full-time roles.

Experience

ML Engineering Intern, FlyRank AI *Jun 2026 - Present*

- Selected for a competitive remote ML engineering cohort from 20,000+ applicants across 100+ countries; collaborating with mentors on production AI systems (embeddings, classification).

Key Projects

NIH Chest X-Ray Multi-Label Disease Classification (Research, 2026)

- Benchmarked 6 architectures on 112K+ chest X-ray images for multi-label classification of 14 thoracic diseases.
- Proposed Swin Transformer V2-B reached 0.8505 mean AUC, surpassing the CheXNet benchmark.
- First author working with academic supervisors; targeting peer-reviewed publication. Stack: PyTorch, Swin Transformer, Grad-CAM.

Synthetic Image Attribution, ICANN 2026 Challenge

- Built a dual-stream deep learning model to identify which of 10 text-to-image models generated a synthetic face image.
- Used two EfficientNet-B4 backbones processing raw RGB and SRM noise-filtered forensic features, fused into a classification head.
- Achieved 95%+ accuracy on the post-processed test set using test-time augmentation and mixed precision training.

Drone Detection in the Wild (YOLO26m + ResNet50, 91.8% mAP)

- Fine-tuned YOLO26m on 1,000+ real-world drone images, achieving 91.8% mAP@50.
- Built a two-stage pipeline combining YOLO26m detection with a frozen ResNet50 classifier as a second-opinion layer.
- Designed a custom augmentation pipeline simulating occlusion, motion blur, and outdoor lighting. Stack: Ultralytics, Keras, Albumentations.

Breast Cancer Classification, Transfer Learning (Best: 98.5% Val Acc)

- Comparative study of ResNet50, VGG16, and EfficientNetB4 for breast cancer ultrasound classification.
- Best model (ResNet50) achieved 98.5% validation accuracy.
- Improved evaluation reliability by removing augmented validation samples to reduce data leakage.

deepcsv, Automatic Data Cleaning Library (PyPI, 13,000+ downloads)

- Open-source Python library for automated CSV/XLSX data cleaning and feature engineering.
- Auto-walks directory trees, converts stringified lists to NumPy arrays, fixes mixed-type columns, and exports to Parquet.
- Added an auto-tuned GradientBoosting mode for automatic feature selection. Install via: pip install deepcsv.

Technical Skills

- **Deep Learning:** TensorFlow, Keras, PyTorch, CNNs, Transfer Learning, custom training loops
- **Computer Vision:** Image classification, object detection (YOLO), OpenCV, ResNet50, VGG16/19, InceptionV3, EfficientNet, Swin Transformer
- **ML & Data Science:** Scikit-learn, Pandas, NumPy, EDA, classification, regression
- **Tools:** Git, GitHub, Jupyter, VS Code, PyPI publishing, Kaggle, Albumentations, Ultralytics

Education

B.Sc. in Management Information Systems (MIS) *2025 - Present*

High Institute for Engineering and Technology, Science Valley Academy, El Obour, Qalyubia, Egypt

Certifications

- Build with AI: Masr Edition, Google for Developers (2026)
- Deep Learning for Computer Vision, MaharaTech / ITI Mooca (2026)
- AWS AI Practitioner Challenge, Udacity (2026)
- AI For Everyone (AI4E), Information Technology Institute (ITI) (2026)
- Python Programming Basics, Information Technology Institute (ITI) (2026)

Languages

Arabic (Native) | English (Intermediate)